

Qubit-Efficient Hybrid Quantum Reinforcement Learning for Scalable Vehicle Routing

T. Satyanarayana Murthy¹

Chaitanya Bharathi Institute of Technology
Hyderabad, India
tsmurthy_it@cbit.ac.in

Santhosh Voruganti³

Chaitanya Bharathi Institute of Technology
Hyderabad, India
vsantosh_it@cbit.ac.in

Chaitanyya Pratap Agarwal⁵

Chaitanya Bharathi Institute of Technology
Hyderabad, India
ugs22034_it.pratap@cbit.org.in

B. Swathi Sowmya²

Chaitanya Bharathi Institute of Technology
Hyderabad, India
swathisowmya_it@cbit.ac.in

Sai Varshini Giridi⁴

Chaitanya Bharathi Institute of Technology
Hyderabad, India
ugs22006_it.varshini@cbit.org.in

Vanteddu Akshitha⁵

Chaitanya Bharathi Institute of Technology
Hyderabad, India
ugs22027_it.akshitha@cbit.org.in

Abstract—The Vehicle Routing Problem (VRP) is a well-known NP-hard optimization problem in logistics and transportation planning. Classical methods face scalability limitations and often rely on static demand and traffic information. Existing quantum approaches, such as the Quantum Approximate Optimization Algorithm (QAOA) and Grover-based search, require deep circuits, high qubit counts, and repeated oracle evaluations, which limits their applicability on noisy intermediate-scale quantum hardware. This work formulates the VRP as a sequential decision-making problem and proposes a hybrid Quantum Reinforcement Learning (QRL) framework. A parameterized quantum circuit is employed as a policy network that learns routing decisions through interaction with the environment and reward feedback. As the problem size increases, Grover Adaptive Search exceeds memory limits and fails to execute, whereas the hybrid QRL method continues to scale efficiently with lower computational and qubit requirements.

Keywords: Vehicle Routing Problem, Quantum Reinforcement Learning, Parameterized Quantum Circuits, Hybrid Quantum-Classical Optimization, Sequential Decision-Making, Noisy intermediate-scale quantum hardware

I. INTRODUCTION

Optimization has always been a crucial part of computational science since it is closely related to making any system more efficient across a wide range of application areas. Along with the use of mathematical precision and the deployment of systematic search strategies, traditional optimization problems have been solved using classical methods such as greedy algorithms, dynamic programming, and branch-and-bound techniques. Nevertheless, with the increasing complexity of real-world systems, which is mostly due to large-scale data, dynamic constraints, and the presence of multiple objectives, classical methods are gradually becoming

practically unfeasible especially when dealing with NP-hard problems. [1]

Greedy algorithms, the earliest optimization methods, are computationally efficient but they generally do not guarantee global optimality. By taking advantage of overlapping sub-problems, dynamic programming can offer better solutions but it quickly runs out of memory because its storage needs grow exponentially. Branch-and-bound tries to improve scalability by pruning but still gets overwhelmed by the exponential growth of the search space as the input size increases [2]. Consequently, metaheuristics such as Genetic Algorithms, Ant Colony Optimization, and Simulated Annealing were introduced to deal with these problems. These methods are scalable and adaptable to various situations, and can generate near-optimal solutions within reasonable time intervals though they sometimes converge slowly or even unstably basically because they depend heavily on randomness and parameter settings [3].

Quantum computing is a new field that exploits the intrinsic quantum mechanical phenomena of superposition and entanglement to explore many possible solutions simultaneously. This quantum feature has prompted the formulation of quantum algorithms designed specifically for optimization problems that require traversing complicated combinatorial landscapes which are far more efficient than their classical counterparts [4]. The Vehicle Routing Problem (VRP), as one of the challenges from the benchmark set, has attracted lots of attention because its solutions directly influence the efficiency of logistics, transportation, and supply chain management. Its combinatorial nature is also the reason why VRP is used as a test case to evaluate classical as well as quantum optimization algorithms [5].

A good illustration of a quantum technique is the Grover

Adaptive Search (GAS) algorithm which is an extension of Grover’s quantum search and the idea is to iteratively amplify the probability of getting low-cost solutions. Specifically, in the VRP, the encoding of feasible routes to quantum states allows the use of amplitude amplification to amplify the probability of measuring the states that represent near-optimal routes [5]. Although GAS has demonstrated that it may have certain computational advantages, it still is a non-adaptive static method. The algorithm neither uses the information obtained from previous iterations for determining the current one, nor does it consider the fact that the routing space may not be completely known beforehand which means that it is very difficult for it to deal with scenarios where the routing environment is both dynamic and uncertain [7].

Quantum Reinforcement Learning (QRL) which is essentially an extension of the classical reinforcement learning technique through integration with quantum computing offers one way of getting around these problems where VRP serves as a case study for network optimization. While classical reinforcement learning is one of the biggest groundbreaking paradigms in the field of artificial intelligence whereby an agent is required to interact with its environment in order to learn sequences of optimal actions, the parallelism capacity of quantum computing has opened new horizons for the extension of these methods. By encoding policies and value functions into quantum states, QRL enables simultaneous exploration of multiple routing strategies and continuous adaptation to environmental changes [6].

While GAS only extends Grover’s fixed probability amplification, QRL utilizes a feedback-driven learning loop to update the control logic. In such a way, the route optimization module can be altered based on the new information coming from the environment. The willingness to embrace variability and fluctuation is, of course, a feature that characterizes the real-world VRP scenarios with constantly changing demands, traffic, and constraints [8]. Besides that, thanks to superposition, QRL may keep a probability distribution over multiple policies which gives it a nice opportunity to achieve a better balance between exploration and exploitation than simple static quantum search methods. Hence, QRL is not only a more flexible and scalable approach to quantum search-integrated learning-based optimization but is the one that can also contribute significantly to the area of quantum intelligent optimization [9].

II. LITERATURE SURVEY

This section summarizes the studies on quantum methods for Vehicle Routing Problem (VRP) and other related combinatorial optimization problems. The research articles have been summarized mainly through four perspectives: (i) quantum annealing and QUBO-based formulations, (ii) variational quantum algorithms such as QAOA, (iii) Grover-based and hybrid quantum search frameworks, and (iv) learning-driven and adaptive quantum optimization methods, including Quantum Reinforcement Learning (QRL). In this review, we constantly draw attention to the major compromises that relate

to the number of qubits, circuit depth, managing constraints, noise susceptibility, and the overall practicality of scaling solutions on near-term quantum devices.

A. Quantum Annealing and QUBO-Based VRP Formulations:

At first, quantum methods for VRP attempted quantum annealing through encoding the routing constraints as Quadratic Unconstrained Binary Optimization (QUBO) problems. The most common encodings use binary variables for vehicle–customer assignments and routing sequences, which makes the problem size rise quadratically. Several papers showed that such models could be run on D-Wave quantum annealers and also considered the extensions of capacitated, multi-depot, and time-constrained VRP variants. These studies verified that routing problems can be physically mapped on quantum hardware, which is a significant proof of the principle.

Meanwhile, real-world issues surfaced quickly. Because of sparse qubit connectivity, expensive minor embeddings are needed which cause a huge increase in the number of qubits used. Moreover, one has to be very careful when setting penalty coefficients to always ensure that feasibility constraints are met without using too much power from the optimization objective. Consequently, most annealing-based methods handle only small-sized problems, and a solution repair process such as local search is usually necessary. Although quantum annealing delivers quick heuristic solutions, due to the limited control that it has over the annealing dynamics as well as hardware constraints, its scalability for real-world VRP applications is very much restricted.

B. Variational Quantum Algorithms for VRP (QAOA and Related Methods):

One of the alternatives to annealing that has received a lot of attention is the use of gate-based variational quantum algorithms, especially the Quantum Approximate Optimization Algorithm (QAOA). This can be done by first expressing the VRP as an Ising or QUBO Hamiltonian, then applying parameterized quantum circuits for optimization with a classical optimizer in the loop. QAOA is a very interesting possibility for Noisy Intermediate-Scale Quantum (NISQ) devices due to its partially classical nature and short circuit requirements.

Experiments and simulations report that QAOA is able to match the performance of classical solvers on small VRP examples. However, a number of issues have to be resolved. The depth of the circuit necessary for getting solutions of good quality increases quite steeply with the size of the problem, which causes the algorithm to become less noise tolerant. In addition, classical optimizers are often at a loss when dealing with noisy or flat cost landscapes, resulting in slow convergence or barren plateaus. Constraint handling is another major difficulty, since the imposition of feasibility usually depends on large penalty terms or complicated problem-specific mixers. Although new research papers are proposing hybrid and warm-start techniques to deal with these problems, the questions of scalable consistent performance and robustness still have to be

explored.

C. Grover-Based Search and Hybrid Quantum Search Frameworks:

Grover-driven quantum search algorithms theoretically give a quadratic speedup for unstructured optimization problems and have been taken up for routing and scheduling problems. When it comes to VRP, such algorithms resort to oracle-based amplitude amplification to increase the probabilities of better routes. Unfortunately, designing efficient oracles that both ensure the feasibility of solutions and at the same time correctly represent cost information is so intricate that very deep quantum circuits are the usual outcome.

There have been proposals of hybrid and multi-stage search schemes in which the feature of separating feasibility determination from cost optimization is explored. Such schemes lower the complexity of the oracle and thus the computational cost and also contribute to a better understanding of the problem from the conceptual point of view. However, their practical effectiveness depends first and foremost on the availability of efficient methods for state preparation and on the precision in iteration control. Although Grover-based methods provide the possibility of an excellent theoretical performance, the depth of the circuits required and the necessity of multiple oracle calls make them almost impossible to realize on currently available noisy hardware. Consequently, most of the work has been theoretical or simulation studies, with very little experimental work.

D. Adaptive and Learning-Based Quantum Optimization Approaches:

Most of the current quantum optimization techniques take VRP to be a static problem that is solved just once and assume that the constraints and demands are totally known beforehand. This assumption thus limits their use in real-life situations where the conditions refer to the routes that can change dynamically. To address this limitation, recent works on research have investigated Quantum Reinforcement Learning (QRL) as an adaptive optimization framework for the problem at hand. QRL equates the VRP to the problem of making a sequence of decisions and a quantum-enhanced agent can learn the routing policies by interacting with the environment and receiving rewards.

QRL takes advantage of quantum superposition to simultaneously explore different routing strategies and at the same time, reinforcement learning carries out policy updates based on the changes in the demands or traffic conditions. Differently from static quantum search methods, the learned policies can be applied to different VRP instances, so that there is less need for re-optimization. In spite of the difficulties related to the quantum state encoding, training stability, and hybrid integration, QRL is a very interesting option for obtaining scalable, flexible, and real-time routing optimization assisted by quantum.

III. PROPOSED METHODOLOGY

A. Grover Adaptive Search (GAS) for Vehicle Routing Problem

In this work, Grover Adaptive Search (GAS) is one of the quantum optimization methods considered for the solution of the Vehicle Routing Problem (VRP). GAS changes Grover's quantum search algorithm so that it can be used for optimization by increasing the amplitude probabilities of quantum states corresponding to solutions with the lowest objective function values [8]. In addition to its ability to use quantum amplitude amplification to explore huge combinatorial spaces, GAS has also been used for routing and other NP-hard optimization problems in the past.

For VRP, any given route plan, which defines the customer-to-vehicle assignment and the sequence of visits, is represented by a computational basis state of a quantum register. The application of Hadamard gates to all the data qubits initializes the system in a uniform superposition of the entire solution space, allowing for the simultaneous evaluation of all the routes [5].

GAS is basically an operation of two unitaries: an oracle and a diffusion operator. The oracle determines the cost of each route and marks the solutions with lower routing cost by a phase flip. The diffusion operator subsequently enhances the probability amplitudes of these marked states. The standard Grover iteration repeats these two operations, with the number of iterations proportional to the square root of the size of the search space.

$$\left\lceil \frac{\pi}{4} \sqrt{\frac{N}{M}} \right\rceil$$

where M is the count of the highlighted (low-cost) solutions.

When the number of Grover iterations has been reached, the quantum state is measured in the computational basis. The bitstring that is observed most often in repeated measurements represents the optimal or approximately optimal route configuration. The index is decoded by a classical post-processing step to obtain the vehicle-to-customer assignments, and the total routing cost is computed [5].

B. Quantum Reinforcement Learning (QRL) for the Vehicle Routing Problem

The proposed approach employs a hybrid quantum-classical reinforcement learning (QRL) agent to solve the Vehicle Routing Problem (VRP). The key idea is to use parameterized quantum circuits as the decision-making component of the agent, enabling efficient exploration of a large combinatorial solution space. Unlike classical or purely quantum search methods, QRL learns from interaction with the environment and can generalize across different problem instances through training. This adaptability makes QRL well suited for scalable and dynamic VRP scenarios [9].

State and Action Representation

At each time step t , the environment state s_t encodes the following components:

- **Vehicle positions:**

$$V(t) = \{v_1, v_2, \dots, v_K\},$$

where v_k denotes the position of the k -th vehicle and K is the total number of vehicles.

- **Customer locations:**

$$C = \{c_1, c_2, \dots, c_N\},$$

where c_i denotes the location of the i -th customer and N is the total number of customers.

- **Visited customer mask:**

$$M(t) \in \{0, 1\}^N,$$

where $M_i(t) = 1$ if customer i has been visited by time t , and $M_i(t) = 0$ otherwise.

The agent's action of choosing at refers to the decision of serving the next customer with a particular vehicle. Formally, a policy is stated as a probability distribution over the set of feasible actions:

$$\pi_\theta(a_t | s_t) = \text{softmax}(\text{QuantumPolicy}(s_t; \theta))$$

where θ stands for the trainable parameters that are shared between the quantum circuit and the classical post-processing layers. The output is normalized so that it becomes a valid probability distribution over all unvisited customers.

1) *Quantum Circuit Encoding:* The agent takes the environmental state s_t and encodes it into a parameterized quantum circuit. Firstly, the state features that consist of vehicle locations as well as the visited customers mask are converted to qubit rotation angles. In a formal manner, the encoding can be represented as

$$\phi_i = f_i(s_t), \quad \forall i \in [1, n_{\text{qubits}}],$$

where each f_i is a conventional preprocessing function that converts the environment state into a rotation angle for qubit q_i . The state of each qubit is given as

$$R_y(\phi_i)|0\rangle = \cos\left(\frac{\phi_i}{2}\right)|0\rangle + \sin\left(\frac{\phi_i}{2}\right)|1\rangle.$$

Entanglement is created by using CNOT gates between adjacent qubits in order to reflect correlations among different vehicles and customers. Several layers of parameterized single-qubit rotations and entangling operations, which are controlled by the parameters θ , are used to increase the expressibility of the quantum circuit. The circuit emits expectation values derived from Pauli-Z measurements on all qubits.

$$\langle Z_i \rangle = \langle \psi(\theta) | Z_i | \psi(\theta) \rangle$$

The method of obtaining these expectation values is through the use of fully connected layers post-processed classically to compute the policy output. $\pi_\theta(a_t | s_t)$.

Action Selection and Reward Computation

At each decision point, the agent uses a hybrid quantum-classical policy network to evaluate a score for each possible vehicle-customer pair. These scores are transformed into a probability distribution using the softmax function, thus actions with a higher expected utility can be selected more frequently while still allowing for stochastic exploration. Once an action (assigning a customer to a vehicle) has been randomly chosen from this distribution, the environment moves to a new state. The reward is calculated by taking the negative Euclidean distance of the selected vehicle's travel. In case an invalid or previously visited customer is selected, a penalty score is added[10].

2) *Reward Function:* The reward at time step t is defined based on the incremental distance traveled by the selected vehicle. It is given by

$$R_t = -\text{dist}(v_t, c_t) - \lambda \cdot \mathbb{I}_{\text{invalid}},$$

During each decision step, an agent scores every feasible vehicle-customer pair through a hybrid quantum-classical policy network. These scores are converted into a probability distribution via the softmax function, which allows actions of higher expected utility to be selected more frequently whilst stochastic exploration is still permitted. Once an action (assigning a customer to a vehicle) is chosen by sampling from the distribution, the environment changes to a new state. The reward is the negative Euclidean distance a selected vehicle travels.

The total accumulated reward, or return, for a trajectory $\tau = (s_0, a_0, R_0, s_1, a_1, R_1, \dots, s_T)$ is defined as

$$G_t = \sum_{k=t}^T R_k,$$

where G_t denotes the return from time step t until the end of the episode at time step T .

3) *Policy Optimization:* Policy gradients methods are used to optimize the policy. Maximizing the expected return is the goal, which is equal to minimizing the loss function below:

$$\mathcal{L}(\theta) = - \sum_{t=0}^T \log \pi_\theta(a_t | s_t) \cdot G_t,$$

The policy is trained by means of the policy gradient method. The main intention here is to maximize the expected return, which is the same as minimizing the loss function given below: Here the notation $\pi_\theta(a_t | s_t)$ depicts the parameterized policy, a_t is the action taken at time step t , and s_t is the corresponding environment state. The parameter update gradient $\nabla_\theta \mathcal{L}(\theta)$, which is a part of gradient descent, is obtained by standard backpropagation for the classical parts and by parameter-shift rules for the quantum gates[11]. Such a hybrid model structure allows transferring of efficiently fine-tuning on larger VRP instances only after initial training, thus

enabling generalization without the necessity of retraining the model from scratch.

Circuit Design

Data qubits denote the encoded features of the environment state such as vehicle positions and the visited customer mask. Parameterized single qubit rotation gates $R_y(\theta_i)$ are used to convert classical input features into quantum probability amplitudes. Entanglement layers consisting of CNOT gates are employed to model dependency relationships between vehicles and customers[12]. Performing measurements on all qubits yields expectation values, which are then passed through classical layers to decide the next action. The layout is so that a handful of qubits are able to represent a large number of cities and vehicles, thus eliminating the problem of one qubit per city, and the system remains scalable.

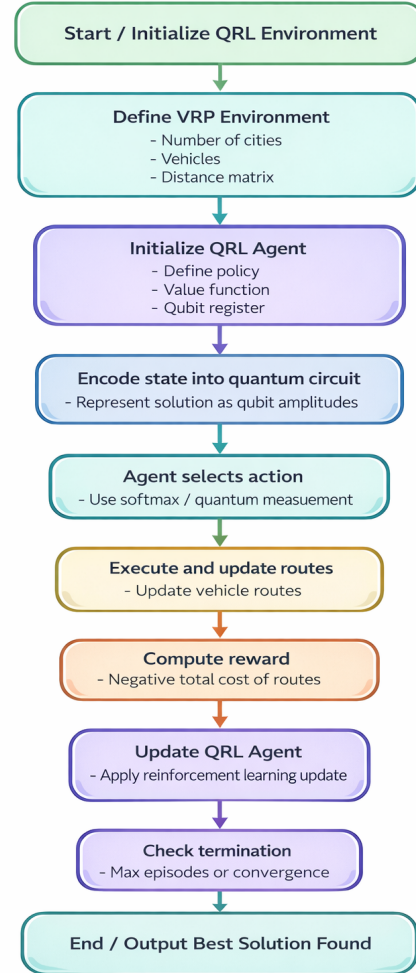
4) *Training and Fine-Tuning*: At first, the agent is trained on small Vehicle Routing Problem (VRP) instances as a means to provide a strong starting point for the quantum policy parameters. After that, the agent is fine-tuned on larger VRP instances, thus, the QRL model can face more difficult problems by itself, without starting over completely from the training[11]. Meanwhile in training, one or more actions are taken from the policy $\pi_\theta(a_t | s_t)$ in order to allow for more random behavior. Rewards are kept for each journey, and then the error which is consequence is used for the backward flow of gradients in order to update the classical and quantum parts of the model through their separate parameter update mechanisms[12].

During training:

- Actions are sampled from the policy $\pi_\theta(a_t | s_t)$ to encourage exploration.
- Rewards are accumulated for each trajectory.
- Gradients are computed to update both classical and quantum parameters[12].

5) *Testing Procedure*: In the test phase, the trained and fine-tuned policy is used on new VRP instances, which was not available during the training. The actions may be determined either greedily (by picking the action with the maximum probability) or randomly by drawing the sample from the probability distribution given by $\pi_\theta(a_t | s_t)$. The generated routes are the vehicle–customer decisions that the quantum circuit most probably produced and hence are the optimized vehicle routes [12]. This section provides:

- formulas for the state, action, reward, and policy;
- details of the quantum circuit design, including rotation and entanglement layers;
- a step-by-step explanation of the hybrid training and fine-tuning procedure;
- an emphasis on scalability, including avoidance of a one-qubit-per-city encoding.



A. Comparative Analysis

a) *Search Complexity*: While Grover Adaptive Search (GAS) theoretically provides a quadratic speedup over a brute-force search, it is exponentially difficult to extend solution space exploration due to dramatic growth of both qubit requirements and circuit complexity. On the other hand, Quantum Reinforcement Learning (QRL) learns a policy distribution and thus does not need to enumerate explicitly, which makes it much less demanding in terms of resources. Despite being a theoretical result, the number of Grover iterations in GAS scales as $\sqrt{N}$. However, the simulation of multi-controlled gates that is necessary for the classical part of the computational load makes the computational cost of the VRP instance simulation rapidly increase and therefore be only limited to small cases[12]. The runtime complexity of the proposed QRL approach scales as

$$\mathcal{O}(E \cdot T \cdot C_{\text{PQC}}),$$

where E denotes the number of training episodes, T represents the number of decision steps per episode, and C_{PQC} is the cost of executing the parameterized quantum circuit.

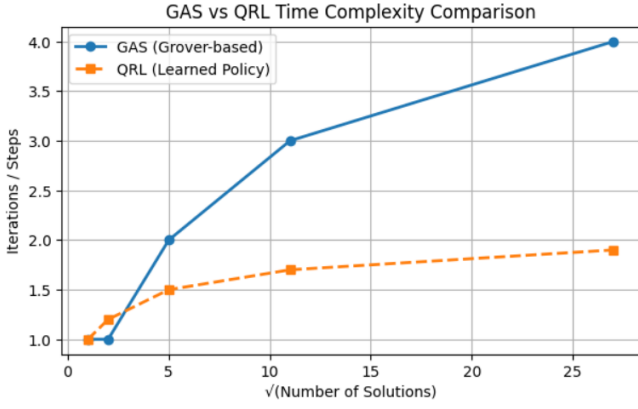


Fig. 1. GAS vs QRL

b) Circuit Metrics: GAS circuits had massive depths and a greater number of multi-controlled gates, both of which increased very inefficiently with the problem size [5]. On the other hand, QRL kept the circuit depth shallow with a fixed number of qubits and variational parameters, hence it was simulable and thus probably runnable on near-term quantum hardware.

c) Memory and Runtime: The memory and CPU time requirements of GAS grew exponentially with the number of solutions and consequently, it was not able to process medium-sized datasets [5]. On the other hand, QRL managed to keep a reasonable memory footprint and runtime even for larger cases, thus showing that it is more efficient in hybrid quantum-classical optimization.

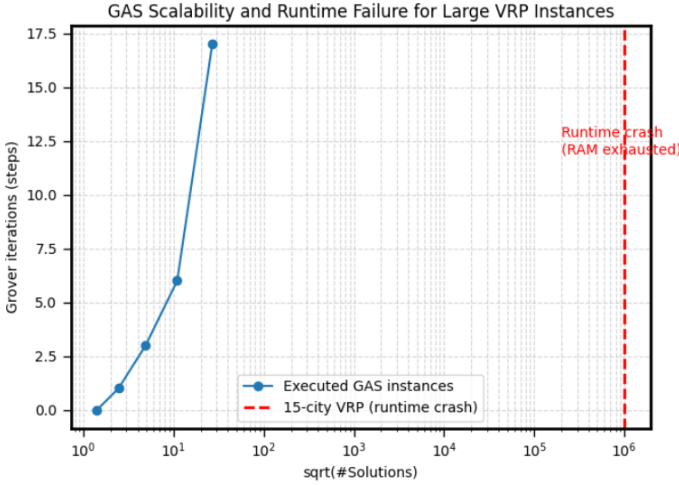


Fig. 2. The 15-city VRP instance failed to execute because the memory requirements were too high. Runtime failure would result from execution exceeding available RAM on Google Colab due to the exponential growth of the VRP solution space and observed GAS scaling for smaller instances.

d) Solution Quality: On small problems both GAS and QRL have found optimal or near-optimal solutions. Nonetheless, QRL was able to take this performance further to bigger instances where GAS was unable to finish the job thus

demonstrating its superior practicability [14]. The method of reinforcement learning facilitated the discovery process in a very efficient manner without having to perform a complete enumeration.

IV. RESULTS

A. Quantum Reinforcement Learning (QRL) Performance

Operations on the data qubits measurement repeatedly resulted in the optimal solution with the highest probability. The measurement histograms coming out from the experiments showed the main peak matching the optimal elements, thus confirming that both the oracle and the Grover diffusion sequence worked correctly. However, the number of qubits currently available and the number of Grover iterations needed limit GAS only for very small problems. Besides, as the number of qubits and the circuit depth grow exponentially, it is impossible to compare GAS with classical or hybrid quantum algorithms on a large scale [14]. The hybrid Quantum Reinforcement Learning (QRL) model was trained on small VRP instances and then fine-tuned on larger datasets to check its capacity for generalization to more complex routing scenarios[11]. In contrast to Grover Adaptive Search (GAS), QRL does not require an exhaustive workspace enumeration but uses a parameterized quantum circuit (PQC) as a policy network combined with reinforcement learning for route optimization[5].

a) Circuit Design: The PQC architecture uses a fixed number of qubits to encode vehicle and customer information. Variational layers composed of parameterized single-qubit rotations and entangling gates allow the model to learn probability distributions over feasible routes without explicit enumeration, supporting efficient gradient-based training.

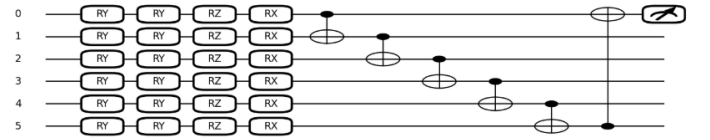


Fig. 3. circuit design

b) Training and Fine-Tuning: The model is trained by generating candidate routes during episodic interactions and receiving rewards based on the total routing distance. Fine-tuning the model with larger VRP instances allows the model to adapt to unseen configurations and enhance route quality without the need to retrain from scratch, thus greatly reducing computational overhead in comparison to GAS.

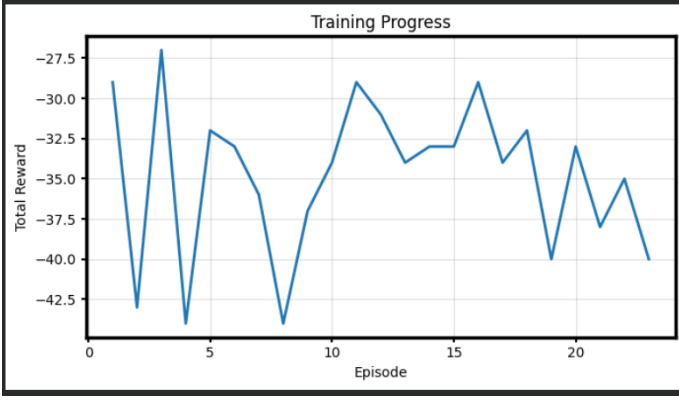


Fig. 4. Negative rewards per episode during QRL training. The plot shows the negative total route cost, with early fluctuations due to exploration.

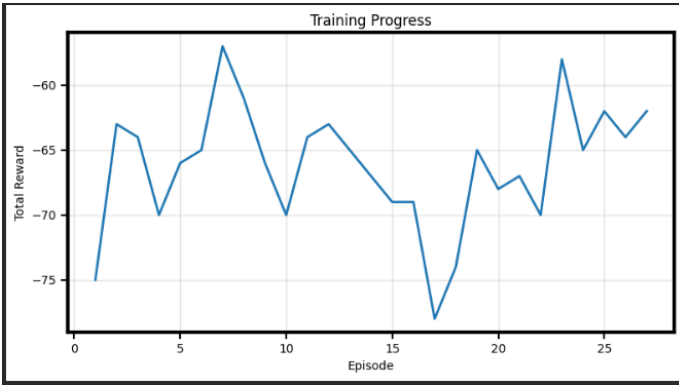


Fig. 5. Negative rewards per episode during fine-tuning.

1) *Evaluation Metrics*: For every test case, several metrics were tracked such as total route distance, number of qubits, circuit depth, and the number of variational parameters[10]. The QRL method produced almost-optimal results for all small and medium scenarios and converged in a stable manner over multiple runs. Besides, memory usage and running time were reduced by several orders of magnitude when compared to GAS for similar problem sizes, which is a consequence of the decreased combinatorial overhead[15].

2) *Scalability*: QRL has proven to be highly scalable on medium-sized problem instances and continues to show reasonable execution times and memory requirements even in cases where GAS runs out of time. Variational optimization steps in the PQC replaced the need for explicit Grover iterations almost entirely, with a convergence that generally happens within a few tens or hundreds of steps depending on the problem complexity[16]. Overall, such findings indicate a strong potential of QRL for a large-scale practical VRP implementation where GAS is out of reach[14].

B. Observations

a) *Scalability*: GAS is appropriate merely for the illustration of minuscule VRP instances since qubit requirements and circuit depth increase very quickly with problem size. On

the other hand, QRL can be scaled effectively to medium-sized instances without the demand for exponentially more resources.

b) *Resource Efficiency*: QRL keeps circuit depth shallow and memory usage low, while GAS circuits become very large and resource-heavy as the problem scales.

c) *Solution Reliability*: Both methods manage to find the optimal or close to optimal solutions on the small problems; however, QRL still keeps producing good solutions in the cases where GAS does not manage to finish.

d) *Time Complexity*: Though GAS theoretically offers a Grover iteration complexity of $O(\sqrt{N})$, it is the classical simulation overhead that takes up most of the runtime, thus in reality, it is slower than the hybrid QRL approach when dealing with medium-sized problems.

V. CONCLUSION

The comparative study shows that GAS quantum algorithm practically offers a clean quantum advantage for small VRP problem instances, however, its practical drawbacks in terms of circuit resources and classical simulation time make its real-world applicability almost zero. The hybrid QRL approach combines quantum circuits with classical reinforcement learning to be able to explore a larger solution space more effectively, hence providing a scalable, memory-efficient, and high-quality optimization technique. Such results serve as a support to QRL being a feasible next step of quantum optimization methods for combinatorial routing problems.

VI. ACKNOWLEDGEMENT

The authors would like to express their gratitude to Dr. T. Satyanarayana Murthy for his valuable guidance and continuous support throughout this research. His advice was crucial in forming the research approach and analysis for the Vehicle Routing Problem (VRP) framework based on Quantum Reinforcement Learning (QRL).

REFERENCES

- [1] R. Martí, P. M. Pardalos, and M. G. Resende, "Handbook of Heuristics," Springer, 2018.
- [2] R. Wang, Z. Zhou, K. Li, T. Zhang, L. Wang, X. Xu, and X. Liao, "Learning to branch in combinatorial optimization with graph pointer networks," IEEE/CAA Journal of Automatica Sinica, vol. 11, no. 1, pp. 157–169, Jan. 2024.
- [3] V. Tomar, M. Rajshree, and P. Singh, "Metaheuristic Algorithms for Optimization: A Brief Review," 2024, doi: 10.3390/engproc2023059238.
- [4] M. Cerezo et al., "Variational Quantum Algorithms," Nature Reviews Physics, 2021.
- [5] L. Liu et al., "Solving Vehicle Routing Problem Using Grover Adaptive Search Algorithm," IEEE Transactions on Intelligent Transportation Systems, vol. 26, no. 7, pp. 9682–9692, July 2025, doi: 10.1109/TITS.2025.3562860.
- [6] A. Alomari and S. Kumar, "A Survey of Quantum Reinforcement Learning Approaches: Current Status and Future Research Directions," 2025, doi: 10.1109/CAI64502.2025.00283.
- [7] H. Ominato, T. Ohyama, and K. Yamaguchi, "Grover Adaptive Search With Fewer Queries," IEEE Access, vol. 12, pp. 74619–74632, 2024, doi: 10.1109/ACCESS.2024.3403200.

- [8] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proc. 28th Annual ACM Symposium on Theory of Computing (STOC)*, 1996, pp. 212–219.
- [9] E. Andrés, M. P. Cuéllar, and G. Navarro, "Efficient Dimensionality Reduction Strategies for Quantum Reinforcement Learning," *IEEE Access*, vol. 11, pp. 104534–104553, 2023.
- [10] B. J. Li, G. H. Wu, Y. M. He, M. F. Fan, and W. Pedrycz, "An overview and experimental study of learning-based optimization algorithms for the vehicle routing problem," *IEEE/CAA Journal of Automatica Sinica*, vol. 9, no. 7, pp. 1115–1138, Jul. 2022, doi: 10.1109/JAS.2022.105677.
- [11] D. T. R. Nagy, A. Plinge, C. Ufrecht, M. Periyasamy, and M. Schuld, "Hybrid quantum-classical reinforcement learning in latent observation spaces," *Quantum Machine Intelligence*, vol. 7, no. 1, pp. 1–15, 2025.
- [12] S. Y.-C. Chen, C.-H. H. Yang, J. Qi, P.-Y. Chen, X. Ma, and H.-S. Goan, "Variational Quantum Circuits for Deep Reinforcement Learning," *IEEE Access*, vol. 8, pp. 141007–141024, 2020, doi: 10.1109/ACCESS.2020.3010470.
- [13] Z. Zhao, A. Anand, and R. Gupta, "Variational Quantum Reinforcement Learning for Sequential Decision Problems," in *Proc. IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2021, pp. 1–8.
- [14] A. Gilliam, S. Woerner, and C. Goniculea, "Grover adaptive search for constrained optimization," *IEEE Transactions on Quantum Engineering*, vol. 2, pp. 1–19, 2021.
- [15] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, "Parameterized quantum circuits as machine learning models," *Quantum Science and Technology*, vol. 4, no. 4, pp. 043001, 2019, doi: 10.1088/2058-9565/ab4eb5.
- [16] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.
- [17] S. Hadfield, "On the Representation of Optimization Problems in Quantum Algorithms," in *Proc. IEEE International Conference on Rebooting Computing (ICRC)*, 2018, pp. 1–8.
- [18] C. Zoufal, R. V. Mishmash, and S. Woerner, "Variational Quantum Reinforcement Learning," in *Proc. IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2020, pp. 1–7.
- [19] H. Yu, M. E. Fouda, and A. E. Youssef, "Quantum-Inspired Reinforcement Learning for Large-Scale Combinatorial Optimization," *IEEE Access*, vol. 9, pp. 118573–118585, 2021.
- [20] J. Niu, J. Chen, and Y. Wang, "Quantum Markov Decision Processes and Reinforcement Learning," *IEEE Transactions on Cybernetics*, vol. 51, no. 4, pp. 1788–1800, Apr. 2021.